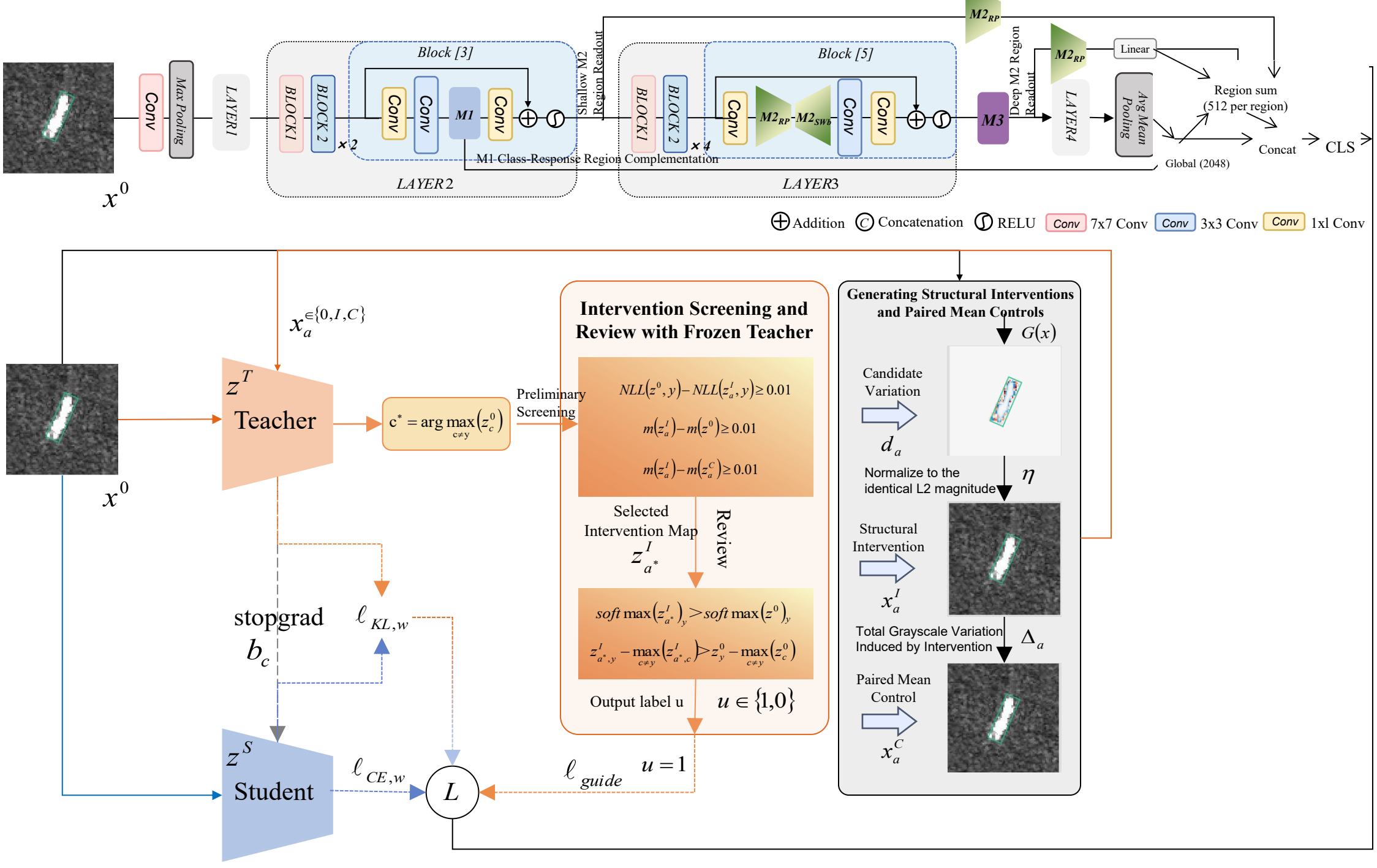

Conv
Max Pooling
LAYER1
BLOCK1
BLOCK 2
× 2
Block [3]
Conv
Conv
M1
Conv
Shallow M2 Region Readout
LAYER2
BLOCK1
BLOCK 2
× 4
Block [5]
Conv
M2RP-M2SWB
Conv
Conv
M3
Deep M2 Region Readout
M2RP
LAYER4
M2RP
Linear
Avg Mean Pooling
Region sum (512 per region)
Concat
CLS
Global (2048)
M1 Class-Response Region Complementation
LAYER3
Addition
Concatenation
RELU
Conv 7x7 Conv
Conv 3x3 Conv
Conv 1xl Conv
Intervention Screening and Review with Frozen Teacher
Generating Structural Interventions and Paired Mean Controls
Teacher
Student
Preliminary Screening
Selected Intervention Map
Review
Output label u
Candidate Variation
Normalize to the identical L2 magnitude
Structural Intervention
Total Grayscale Variation Induced by Intervention
Paired Mean Control
stopgrad